# Statistical analysis of differentiable sliced optimal transport plans

Optimal transport (OT) matches two distributions through a transport *plan*; recent "sliced" methods approximate this plan cheaply by optimizing one-dimensional projections fitted to a finite sample. Whether such a fitted plan stays close to the true OT plan on unseen data — its *generalization* — is unknown: no bound exists. This internship aims to derive the first one, using PAC-Bayes (a learning-theory tool for averaged models) as the entry point and building up, in stages, to a guarantee on the plan itself.

## §1. Organization

- **Start date**: As soon as possible (flexible).
- **Localization**: IRISA, Rennes.
- **Supervision**:
  - Romain Tavenard (Prof., Université Rennes 2, MALT Inria team, IRISA, Rennes).
  - Paul Viallard (MALT Inria, IRISA, Rennes).

## §2. Context

Optimal transport (OT) compares two probability distributions by computing the cheapest way to move the mass of one onto the other. It returns two things at once: a number, the *distance* (how costly the move is), and a *plan* (which mass goes where). The plan is what makes OT so useful in practice — for matching, alignment, domain adaptation, or generative modelling — but computing it exactly becomes very expensive as the dimension grows.

*Sliced* OT is a popular way around this cost. The idea is to project the two distributions onto random one-dimensional lines, where OT has a cheap, closed-form solution, and then to combine the results across many projections. Averaging the one-dimensional *distances* over random projections gives a fast and reliable surrogate for the OT distance; this averaged form is what most of the existing theory studies.

Recovering a *plan* from slicing is more recent and more delicate. DGSWP (Chapel et al., 2025) does not merely average one-dimensional distances: for a given projection it lifts the one-dimensional plan back to the original space and measures its *true, full-dimensional* transport cost, and it then *learns* the projection that makes this cost as small as possible. The output is an approximate OT *plan* in the original space, obtained by optimizing a projection on the data at hand.

Fitting a plan to data raises a *generalization* question, central in machine learning: a plan that fits the available samples well may still poorly reflect the true underlying distributions. We would like a guarantee that the fitted plan is close to the *true* OT plan. The natural tool is PAC-Bayes, a branch of learning theory that bounds the gap between performance on the sample and performance on the (unseen) population whenever the quantity of interest is an *average over a distribution of models* — here, an average over projection directions. Ohana et al. (2023) did exactly this for the sliced *distance*, even learning the best distribution of projections by optimizing the bound, and the supporting statistics of sliced distances are well developed (Nietert et al., 2022). No comparable guarantee exists for a sliced *plan*, which is harder because it lives in the full space rather than on the line.

## §3. Scientific challenge

The goal is a bound that certifies how well a DGSWP plan, fitted on a finite sample, reflects the true OT plan between the underlying distributions. A natural route reaches it in three steps of increasing difficulty.

The first step adapts the PAC-Bayes analysis of Ohana et al. (2023) to our estimator. DGSWP also averages a cost over a (learned) distribution of projections, so its objective again has the "average over models" form that PAC-Bayes needs. The difference is what is averaged: a full-dimensional plan cost rather than a one-dimensional distance. This yields a bound on how far the value measured on the sample is from its true, population value.

The second step connects that value to true OT, by separating two effects that are easy to conflate. One is a *smoothing* gap: to make the projection learnable by gradient descent, DGSWP optimizes a smoothed version of the ideal objective, and one must measure how much this smoothing costs. The other is a *slicing* gap: even with no smoothing, the best single projection only approximates the true OT plan. Quantifying each gap, and how they grow with dimension, is at the core of the internship.

The third step moves from the transport *cost* to the transport *plan* itself — the object we ultimately care about. The natural tool is *quantitative stability* of OT, a body of results showing that two transport problems with similar costs must have similar plans (Kitagawa et al., 2019; Bansil & Kitagawa, 2022); it has just been used to analyze an averaged OT plan (Boïté et al., 2026), which closely echoes DGSWP, whose plan is itself an average. These results are not yet stated for a lifted sliced plan, so adapting them is the final and most exploratory part.

## §4. Goals

Concretely, the candidate will carry out the three steps above and validate them empirically against the true error on controlled distributions where the exact OT plan is known.

Candidates should possess:

- a solid probability and statistics background,
- the ability to read PAC-Bayes-style proofs,
- good Python coding skills for experiments,
- ability to communicate in English in a scientific context.

Funding for a Ph.D. has already been secured: upon a successful internship, the candidate will be offered the opportunity to continue this work as a fully funded doctoral thesis. A successful candidate will develop or reinforce skills key to that continuation, namely:

- a practical and theoretical command of generalization theory (PAC-Bayes and beyond),
- familiarity with the statistical and computational properties of sliced optimal transport,
- experience turning a theoretical bound into a rigorous experimental protocol.

## §5. Bibliography


Alquier, P. (2024). *User-friendly Introduction to PAC-Bayes Bounds.* Foundations and Trends in Machine Learning.

Bansil, M., & Kitagawa, J. (2022). *Quantitative Stability in the Geometry of Semi-discrete Optimal Transport.* International Mathematics Research Notices.

Boïté, S., Delon, J., & Nadjahi, K. (2026). *Expected Batch Optimal Transport Plans and Consequences for Flow Matching.* arXiv:2605.12174.

Kitagawa, J., Mérigot, Q., & Thibert, B. (2019). *Quantitative Stability of Optimal Transport Maps and Linearization of the 2-Wasserstein Space.* AISTATS.

Nietert, S., Sadhu, R., Goldfeld, Z., & Kato, K. (2022). *Statistical, Robustness, and Computational Guarantees for Sliced Wasserstein Distances.* NeurIPS.

Ohana, R., Nadjahi, K., Rakotomamonjy, A., & Ralaivola, L. (2023). *Shedding a PAC-Bayesian Light on Adaptive Sliced-Wasserstein Distances.* ICML.

Chapel, L., Tavenard, R., & Vaiter, S. (2025). *Differentiable Generalized Sliced Wasserstein Plans.* NeurIPS. arXiv:2505.22049.